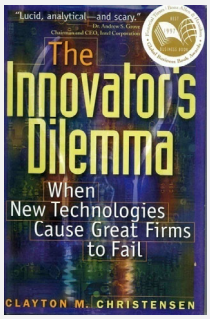
EAS5450 Class #18

The entrepreneurial opportunity of disruptive technology.



Penn Engineering

1

Creative Destruction

"Entrepreneurship is a process of creative destruction; one that incessantly revolutionizes the economic system from within, incessantly destroying the old one, incessantly creating a new one. This process of Creative Destruction is the essential fact about capitalism."

- Joseph Schumpeter, 1934

2

M.B.A. Maxims…

- Stay close to your customer!
  - *The customer is always right.*
  - *Spend a day in the life of your customer.*
  - *Customer-driven product development.*
- Create barriers to entry.

3

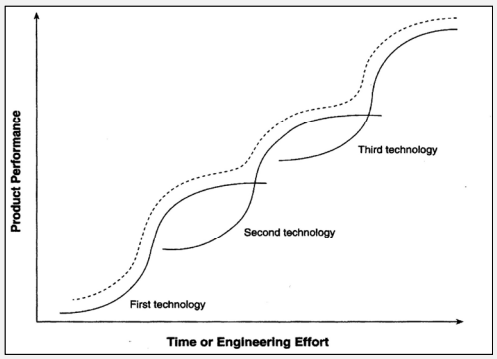
Sustaining Technologies

… evolve along performance parameters historically valued by customers.

TECHNOLOGY	PERFORMANCE PARAMETER
Computer Disk Drives	Storage Capacity

4

Sustaining Technology S-Curves



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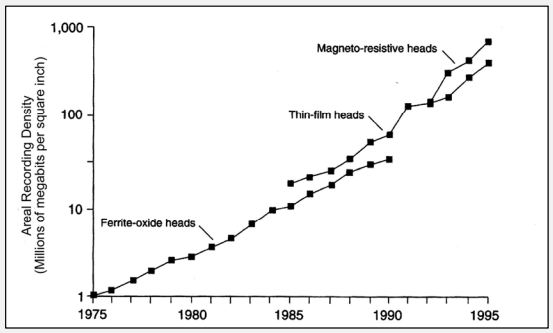
Sustaining Technology Trajectory

Computer Disk Drive Technology



6

The Cost of Staying Close

	31-Dec-08	
	$1,000's	% of Rev
Revenue	**$14,876.16**	**100%**
Cost of Revenue, Total	$6,653.79	45%
Gross Profit	**$8,222.37**	**55%**
Selling/General/Admin. Expenses, Total	$4,601.59	31%
Research & Development	$1,721.33	12%
Unusual Expense (Income)	$330.51	2%
Total Operating Expense	**$13,307.23**	**89%**
Operating Income	$1,568.94	11%
Other, Net	($39.41)	**0%**
Income Before Tax	$1,702.81	11%
Income After Tax	$1,390.29	9%
Minority Interest	($44.73)	0%
Net Income	**$1,345.57**	**9%**

EMC Corp

Source: Hoovers, EMC 10-K 2008

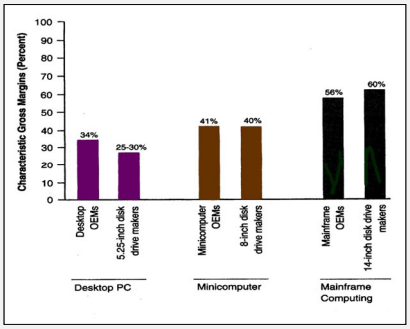
Gross Margin = 55%

R&D = 12%

Sales, G&A = 31%

7

Gross Margin Comparisons



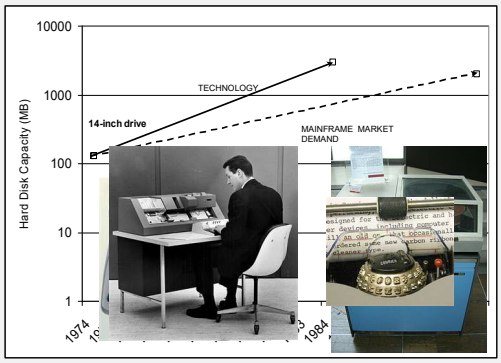
8

Staying Close to the Customer…

- Allocates resources to attack up market:
  - Higher performance
  - Higher margins.
- Precludes downward mobility.
- Defines and delimits what companies can and cannot do.

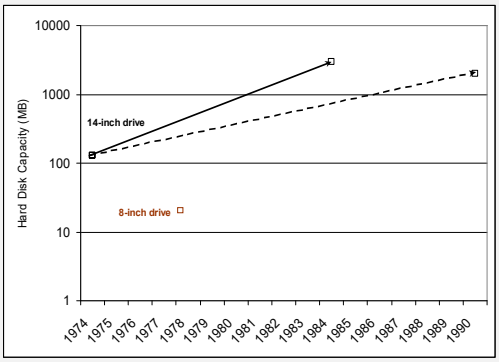
9

Disk Drive Technology



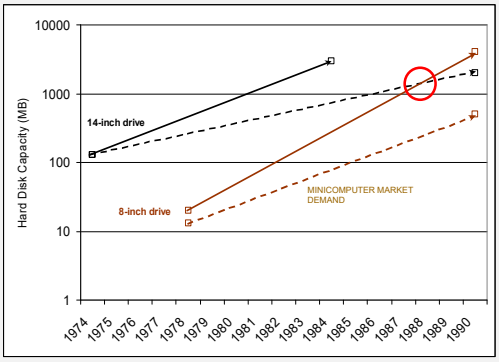
10

Disk Drive Technology



11

Disk Drive Technology



12

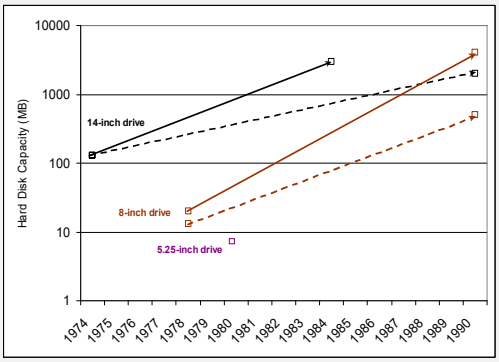
"Disruptive Technology"

…expands the value proposition.

- Adds new product attributes to the value proposition.
- Attributes initially valued only in emerging or low-end markets, unimportant to the mainstream.
- Not simply improvements to attributes already part of the value proposition.

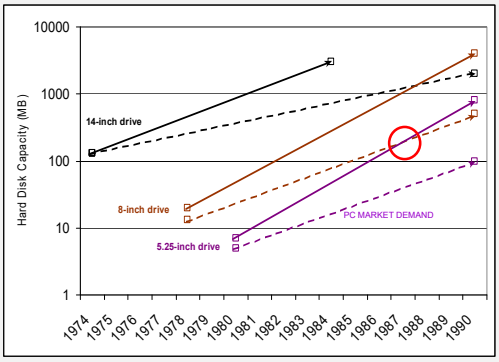
13

Disk Drive Technology



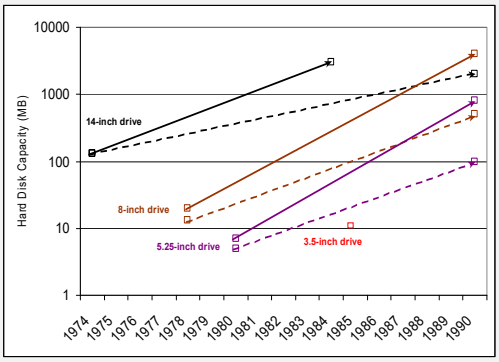
14

Disk Drive Technology



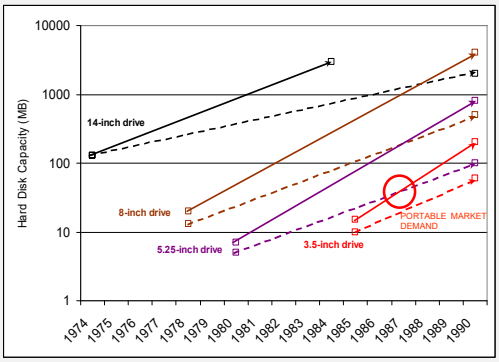
15

Disk Drive Technology



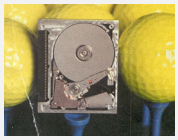
16

Disk Drive Technology



17

Sustaining or Disruptive?



IBM Micro-Drive

Flash Drive

Toshiba 0.85" HDD

Thumb Drive

18

Solid State Disk Drives

- Internal SSDs
- 512GB
- 2.5 inch
- 550 MB/sec read
- 500 MB/sec write

- Samsung PM1643
- 30.72TB
- 2.5 inch
- 2,100 MB/sec read
- 1,700 MB/sec write

19

Sustaining or Disruptive?

UBER

iPhone

20

Corporate Disadvantage

- Established companies focus up market:
  - Cost structures require high margins
  - Emerging market risk perception
  - Captive to mainstream customers
  - Competencies tied to existing lines
  - Managers' personal credibility at stake
- No downward vision or mobility

21

Mature Corporations

- Bound by their assets
- Bound by their infrastructures
- Bound by their business models
- Bound by their expertise

22

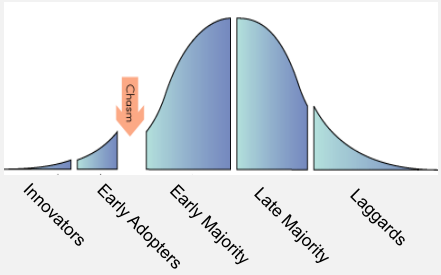
Entrepreneurial Opportunity

Start-Ups and Spin-Offs

- Match the size of the company to the size of the market.
- Nimbler, first mover advantage:
  - First in emerging market,
  - Then in mainstream market.
- Lower cost structure:
  - Cost leader advantages
  - Profitability at lower pricing.

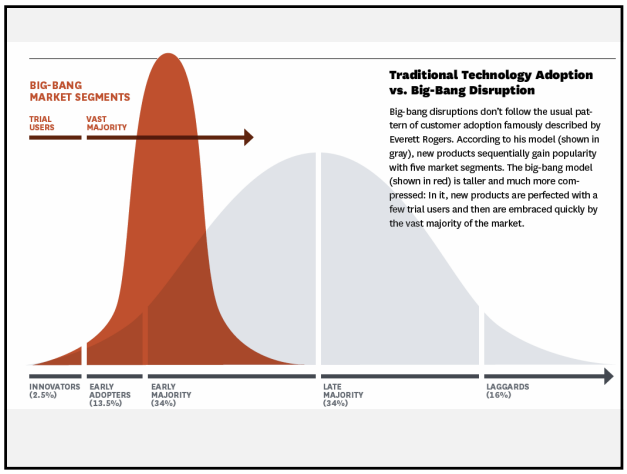
23

Technology Adoption Life Cycle



- G.A.Moore, "Crossing the Chasm"

24



25

Time Value of Money

Would you rather receive:

- $1000 today, or
- $1000 in 5 years?

26

- There is an "opportunity cost" of money:
  - What return could we earn from our best alternative investment opportunity?
  - What interest (dividend) rate must we pay on money we borrow (raise)?

27

Future Value Analysis

Year	Amount at Beginning of Year	Interest Earned During the Year	Amount at End of Year
1	P	Pi	P + Pi = P(1+i)
2	P(1+i)	P(1+i)i	P(1+i) + P(1+i)i = P(1+i)(1+i) = $P(1+i)^2$
3	$P(1+i)^2$	$P(1+i)^2 i$	$P(1+i)^2 + P(1+i)^2 i = P(1+i)^2(1+i) = P(1+i)^3$
⋮	⋮	⋮	⋮
n	$P(1+i)^{n-1}$	$P(1+i)^{n-1} i$	$P(1+i)^{n-1} + P(1+i)^{n-1} i = P(1+i)^{n-1}(1+i) = P(1+i)^n$

$F = P\,(1+i)^n$

28

Example: Future Value

$F = P\,(1+i)^n$

- P = $1,000 today.
- Invested in securities yielding: i = 15% p.a.
- What is the value after n = 5 years?

$F_5 = \$1{,}000\,(1.15)^5 = \$2{,}011$

29

Present Value Analysis

$F = P\,(1+i)^n$
therefore…
$P = F\,(1+i)^{-n}$

30

Net Present Value Analysis

Opportunity Cost of Capital = 12% p.a.

?

Year	Alternative "A"	Alternative "B"	Discount Factor (1.12^n)	Alternative "A" NPV	Alternative "B" NPV
0	-$50,000	-$60,000	1.000	-$50,000	-$60,000
1	7,000	7,000	0.893	6,250	6,250
2	7,000	7,350	0.797	5,580	5,859
3	7,000	7,718	0.712	4,982	5,493
4	7,000	8,103	0.636	4,449	5,150
5	7,000	8,509	0.567	3,972	4,828
6	7,000	8,934	0.507	3,546	4,526
7	7,000	9,381	0.452	3,166	4,243
8	7,000	9,850	0.404	2,827	3,978
Σ	$6,000	$6,844		-$15,227	-$19,672

31

Net Present Value Analysis = f(i)

Year	Actual Cash Flow	6% Discount Rate		9% Discount Rate		12% Discount Rate	
		Factor	NPV	Factor	NPV	Factor	NPV
0 (today)	-$50,000	1.000	-$50,000	1.000	-50,000	1.000	-50,000
1	9,000	0.943	8,491	0.917	8,257	0.893	8,036
2	9,000	0.890	8,010	0.842	7,575	0.797	7,175
3	9,000	0.840	7,557	0.772	6,950	0.712	6,406
4	9,000	0.792	7,129	0.708	6,376	0.636	5,720
5	9,000	0.747	6,725	0.650	5,849	0.567	5,107
6	9,000	0.705	6,345	0.596	5,366	0.507	4,560
7	9,000	0.665	5,986	0.547	4,923	0.452	4,071
8	9,000	0.627	5,647	0.502	4,517	0.404	3,635
	$22,000		**$5,888**		**-$187**		**-$5,291**

32

GenapSys

EAS5450:
Engineering Entrepreneurship I

Penn Engineering

33

GenapSys GENIUS - DNA Sequencer

GENIUS - DNA Sequencer
- Low cost
- Small size
- Easy to use
- Fast

Strategy ?

Team
- Hesaam Esfandyarpour, Ph.D.
- Leila Rastegar
- Engineers
- Distinguished SAB

Cash
- 2010-2013: $4M NIH Grants
- April 2013: $10M Series A
- Late 2013: $20-40M Series B (planned for product release)
- Burn rate: ~$700,000/mo

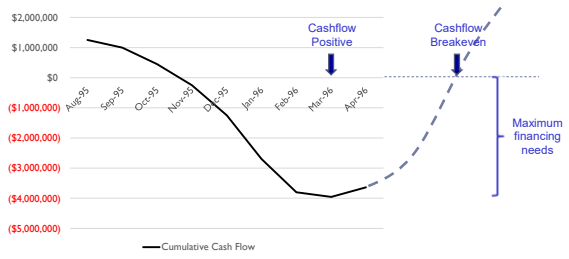
34

GenapSys GENIUS – Business Models

- **Traditional ?**
- **Razor-and-Blade ?**
- **Data Analytics ?**

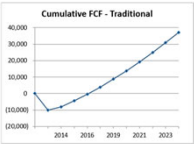
35

Cumulative Cash Flow Determines Funding Needs

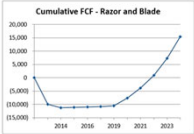


36

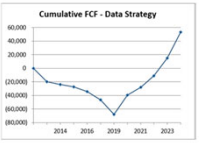
Note different Y-Axis Scales ($ Millions)



Maximum Financing Need = $10M
Risk-Adjusted Discount Rate = 12%
Net Present Value = $37,751,000



Maximum Financing Need = $11.2M
Risk-Adjusted Discount Rate = 15%
Net Present Value = $35,908,000



Maximum Financing Need = $67.9M
Risk-Adjusted Discount Rate = 20%
Net Present Value = $49,821,000

37

GenapSys GENIUS – Business Models

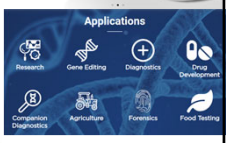
- **Traditional ?**
- **Razor-and-Blade ?**
- **Data Analytics ?**

38

42

Epilogue: GenapSys

- Pursued Razor-and-Blade model with focus on
  - Cancer diagnostics
  - Rare disease
    Signed partnership with Twist Biosciences to create a sequencing workflow for rare disease diagnostics, 2020.
  - Synthetic biology
  - Global DNA Read, Write and Edit Market
  - Liquid Biopsy
    GENIUS co-marketed by Signma Aldrich (subsidiary of Merck KGAA)



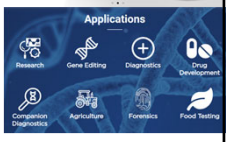
Streamlined 4-step workflow using the GenapSys Platform

Penn Engineering

43

GenapSys

- GenapSys completes product across wide variety of sequencing applications
  - GenapSys Sequencer
  - GenapSys Sequencing Prep System to automate clonal amplification & enrichment
  - Sequencing Kits with reagents & consumables (CMOS chips)
  - Data Analysis output to standard FASTQ file and GenHub hosted solution
- $320MM raised to date (via PitchBook)
  - $75MM debt funding (Feb 2020)
  - Plus, in April 2020, PPP loan to retain 92 employees
  - May 2021: $70M Series D equity funding



Streamlined 4-step workflow using the GenapSys Platform

Penn Engineering

44

- Genapsys names Jason Myers as CEO replacing founder, Hesaam Esfandyarpour, effective Feb 15, 2021.

  www.genomeweb.com: 2/3/2021

- Genapsys lays off one-third of workforce less than one year after raising $70M Series D.

BANKRUPTCY

Venture-Backed DNA Sequencing Firm GenapSys Files for Bankruptcy

The venture capital-backed company says mounting litigation costs have contributed to its strained finances

Wall Street Journal: 7/12/2022

- Court approves sale of Genapsys to Sequencing Health for $10M cash and assumption of $32M debt.

  www.genomeweb.com: 9/12/2022

45